

Brandon Luo

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EDUCATION

Georgia Institute of Technology | Electrical Engineering B.S. | 4.0 GPA

Expected Graduation: May 2028

Stamps President's Scholar

- Selected as the **top 0.01% of Georgia Tech** (1 of 50 scholars) and awarded a **full-ride merit scholarship**
- Relevant coursework: **Digital System Design**, Circuit Analysis, Programming HW/SW Systems (Assembly/C++)

TECHNICAL SKILLS

PCB & Circuitry: Ethernet, mixed-signal circuits/layout, tolerance/limit analysis, CAN FD, I2C, SPI, UART, ARINC-429, RS-422

Tools: Altium Designer, KiCAD, Siemens NX, Git, oscilloscopes, waveform generators, multimeters

Programming/Software: Python, C++, Java, MATLAB, LabVIEW, SystemVerilog, RISC-V

INTERNSHIPS

BETA Technologies | Flight Computer and Controls

May 2026 – August 2026

PCB Design Intern

- Designed circuits and PCB layouts in Altium and KiCAD for **galvanically isolated** dissimilar flight computer modules (Angle of Attack sensors, Weight on Wheels sensors, and Ethernet/CAN FD/RS-422/ARINC 429 transceivers)
- Designed a **Gigabit Ethernet** transceiver circuit through PHY specification and characteristic impedance analysis
- Ran **LTspice Monte Carlo simulations** to analyze component drift and RSS worst-case tolerances
- Achieved **0.2 degrees AOA accuracy** over a range of 80 degrees using unity gain buffers and RC filters, accounting for component and environment tolerances; built and tested 3-microsecond **high-side overcurrent protection**
- Designed an **interface board** to test flight computers and avionics units: AHRS, Garmin displays, Speedgoat test racks
- Wrote and released **PCB fabrication/assembly drawings** for battery pack components, submitted to FAA certification

SpaceX | Starlink Avionics

January 2026 – April 2026

Avionics Engineering Intern

- Validated Starlink V3 PCBs** for flight readiness via in-circuit and functional load testing, establishing tolerances
- Used Siemens NX to design/route copper and fiber optic harnesses for power and data, **flown on Starship Flight 13**
- Developed an **automated micron-accurate fiber termination** machine using LabVIEW, C#/.NET, and RS-232; implemented edge detection and a dedicated GUI to achieve a **2x production efficiency**

INVOLVEMENTS

Digital Design Engineer | SiliconJackets

August 2026 – Present

SiliconJackets is a student-led semiconductor engineering team that designs, verifies, and tapes out chips

- Designed a processor in **SystemVerilog**, implementing datapath, control FSM, register file, ALU, and SRAM interfaces
- Implemented decode and execution for core instructions (add, sub, lw, sw, beq, shift, ebreak)
- Verified RTL** via simulation, validating **SRAM timing**, instruction sequencing, and register states

Battery Management RE | Yellow Jacket Space Program

August 2025 – Present

The Yellow Jacket Space Program builds, tests, and launches liquid rockets with the goal of reaching space in May 2027

- Lead engineer for power distribution, designing and simulating **240W load switches** and **active cell balancing**
- Implemented **MOSFET logic** for power source selection and alternation, particularly during fault conditions
- Conducted extensive field integration and testing, troubleshooting continuity faults and signal integrity issues

Peer Instructor | The Hive Makerspace

February 2026 – Present

The Hive is an EE makerspace for students. It contains benchtop instruments, PCB manufacturing machines, and 3D printers

- Certified in the advanced operation of **benchtop instruments** (oscilloscopes, waveform generators, power supplies)
- Guided students in the use of PCB manufacturing machines, laser cutters, and bench instruments for their projects

PERSONAL PROJECTS

Thrust Vectoring Monocopter Drone

June 2025 – February 2026

- Wrote a **PID control system** using **IMU feedback** to counter angular momentum for dynamic attitude stabilization
- C++, Teensy 4.0, MPU6050**, HC-05 Bluetooth module, RS2205 motor, 4s lipo battery, servos, custom 3D printed parts